

Name: _____

Class: _____

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**CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION****ANSWERS****BIOLOGY****Paper 3 Applications Paper****9747/03****Wednesday 10 September 2008****1 hour 30 min**Additional materials: Answer booklet
Writing Paper**READ THESE INSTRUCTIONS FIRST**

Write your name and HT group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Show All working

At the end of the examination, fasten all your work securely together;

The number of marks is given in brackets []
at the end of each question or part question.

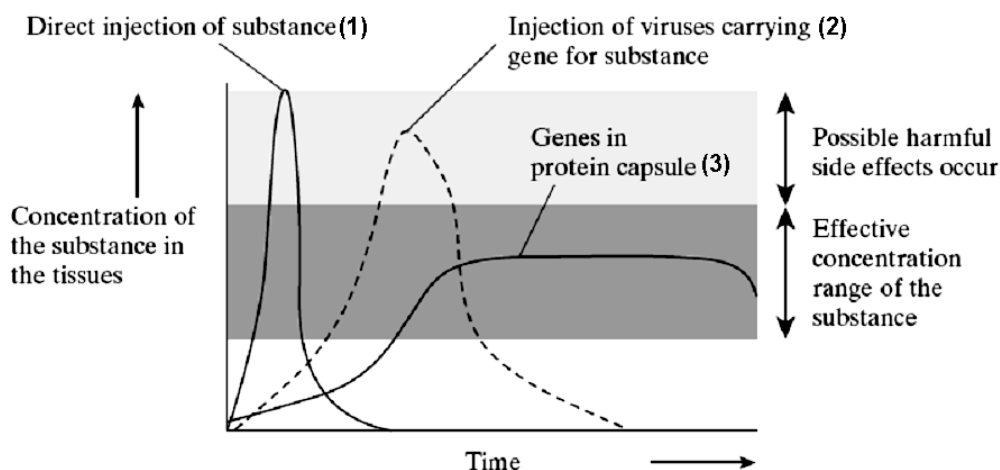
For Examiner's Use	
1 [15]	
2 [15]	
3 [15]	
4 [20]	
Total	65

This document consists of **9** printed pages.**[Turn over]**

- 1 In gene therapy, genes are introduced into a person who has defective genes which do not produce an important substance. Three experiments were done to compare techniques for introducing an important substance into a person with defective genes.

1. The substance was injected directly.
2. Harmless viruses carrying genes coding for the substance were injected.
3. The genes were put into a protein capsule which was inserted into the tissues.

The graph shows results of the experiments.



- (a) (i) State three key differences between graphs 2 and 3. [3]

1. **Gradient for the increase** in substance is steeper in graph 2 then in 3
2. **increase in substance** is faster in graph 2 then in 3
3. In 3, amount of time that the **concentration is in the effective range** is **4.5 times** that of 2

OR

1. In 3, amount of time that **the concentration is in the effective range** is **greater than in 2.**
2. **Concentration of substance reaches toxic concentrations** (harmful side effects occur) but in **2 but not in 3.** Concentration of substance remains below toxic levels in 3.
3. **Duration where substance remains in body is longer in 3 than in 2**

- (ii) Using the information in the graph, suggest **two** advantages and **one** disadvantage of the capsule method compared to the others. [3]

Advantages

1. **substance never reaches harmful levels/no side effect/less likely to harm the organism;**
2. **longer relief from symptoms/ longer effective range/less frequent treatment needed**

Disadvantage

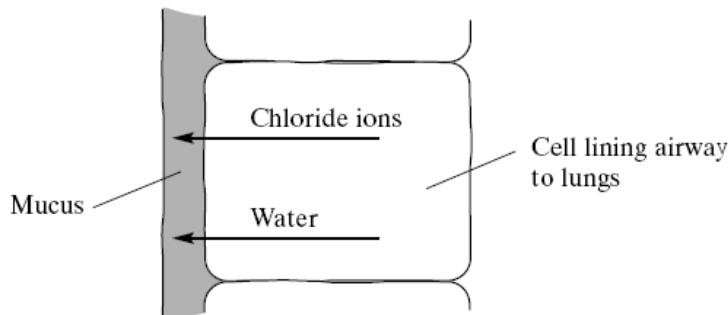
3. **takes longer to take effect**

- (b) (i) Some people are concerned about using gene therapy to treat genetic disorders in humans. Give **three** arguments against the treatment of disorders by gene therapy. [3]
1. **high cost** plus reason e.g. compared with conventional treatments
 2. risk of **stimulating the immune system** (by the vector)
 3. use of **animals in (preliminary) testing**, other genes introduced which may have **damaging effects**/damage genome/damage other genes
 4. may induce **a tumor if integrated in a tumor suppressor gene** because insertional mutagenesis → cancer may result.
 5. **short-lived nature** of gene therapy so patients **will have to undergo multiple rounds** of gene therapy
 6. **immune system's enhanced response** to invaders it has seen before makes it **difficult for gene therapy to be repeated in patients**
 7. possibility of **viral vector recovering its ability to cause disease** once in host
 8. **inadequate understanding** of biological interactions of vector and host
 9. conditions or disorders that arise from mutations in a single gene are the best candidates for gene therapy, but some the most commonly occurring disorders are caused by the **combined effects of variations** in many genes. **Multigene disorders especially difficult to treat effectively using gene therapy**
 10. changes to genetic make-up of individual/species/genome/future generations/germ line may affect normal development of offspring

Reject - not ethical (without explanation)

- (ii) Explain why a person cured by gene therapy may still have children who suffer from the abnormality. [2]
1. **gene therapy treats somatic cells and gametes/germ cells are not affected/do not take up the healthy gene**
 2. **will still pass on the defective genes**

- (c) The diagram shows the movement of chloride ions and water in the cells of the lung of a healthy human.



- (i) In people with cystic fibrosis the gene controlling the movement of chloride ions is defective. The defective gene produces defective CFTR. Suggest why defective CFTR does not transfer chloride ions across cell membranes. [1]
1. **defective CFTR has different tertiary structure / conformation / different shape so channel blocked / chloride ions do not fit**
- (ii) People with cystic fibrosis produce mucus which is stickier and thicker than healthy people. Suggest why people with cystic fibrosis are more likely to develop bacterial infections, such as pneumonia. [1]
1. **mucus not removed so mucus traps bacteria allowing more time for bacteria to breed**
- (iii) Describe briefly **one** way in which normal CFTR genes might be introduced into the cells lining the airways of the lungs. [2]
1. **Liposomes: Functional CFTR gene into liposomes directly / into plasmids into liposomes;**
 2. **Use of aerosol spray / inhalers to deliver vector**
 3. **liposomes fuse with membrane of cells delivering genes across membrane into cells**

OR

1. Virus: Functional CFTR gene into attenuated virus
2. Use of aerosol/sprays/inhalers to deliver vector
3. Virus infects cells delivering genes across membrane into cells

[Total : 15]

2 Restriction Enzymes (also known as restriction endonucleases) are used to cut DNA molecules.

(a) State the source and natural use of restriction enzymes.

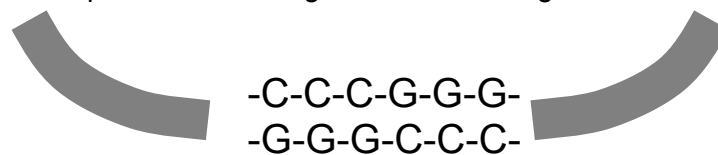
[2]

1. **produced by bacteria** as a natural **defence mechanism**
2. to **breakdown DNA** of invading **viruses and phages**.

The target sites for three restriction enzymes are shown in Table 2.1.

Table 2.1	
Name of restriction enzyme	DNA base sequence of target site
A	<pre> -C-C-C G-G-G- -G-G-G C-C-C- </pre>
B	<pre> -C- C-G-G- -G-G-C- C- </pre>
C	<pre> -A- A-G-G-T-T- -T-T-C-G-A- A- </pre>

Fig 2.1 shows part of a plasmid, including details of the target site for restriction enzyme A.



(b) Explain

(i) whether it would also be possible to cut this sequence using restriction enzymes B and C. [2]

enzyme B

1. Yes, target site is present (even though it will produce sticky ends)

enzyme C

1. No, Target site absent

(ii) why restriction enzymes will only act at certain target sites within the DNA molecule. [3]

1. The active sites have a **specific configuration**
2. different base pairs **have different shapes**
3. there is a **critical need for both to be complementary** -ref to affinity to active site.

- (c) Plasmids are loops of DNA which are found in bacteria. They can be used to transfer genes into bacterial cells. In some cases, these may be human genes.
- (i) Describe the steps that need to be taken in order to insert a human gene into a plasmid. [4]
1. **Specificity** of the restriction enzyme cutting only at **restriction sites** (palindromic sequences)
 2. the **human gene of interest** must be cut with the **same restriction enzyme** thus producing **complementary stick ends**
 3. these sticky ends **complementary base pair according to Watson and Crick base pairing** formation of **hydrogen bonds** between complementary bases.
 4. **action of Ligase**
- (ii) Using a named example, explain the advantages of being able to transfer human genes into bacteria. [4]
1. Incorporation of human genes within bacteria **will allow production of human proteins within bacteria.**
 2. as these are **exactly the same as those produced in humans**, it will **not stimulate an immune response.**
 3. e.g. **insulin / human growth hormone / somatotrophin**
 4. **reduces the need for external sources** of protein.
 5. these can be **mass produced in large quantities.**
 6. **speed in production** of these proteins
 7. **overcoming ethical objections** (compared to production in live animals)

Max 4

[Total 15]

- 3 Equal masses of tobacco plant callus were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin.

The results are shown in Table 3.1.

treatment	concentration of plant growth regulators / mg dm ⁻³		effect of plant growth regulators on callus growth
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

- (a) Explain why growing plantlets from a callus in tissue culture results in a clone. [3]
1. *This is a type of asexual reproduction.*
 2. *All the plantlets are genetically identical / have the same genotype because they are produced by mitotic division.*
 3. *The process of mitosis involves exact replication of the parental DNA.*
- (b) State why the plant tissue cultures are initially grown in sterile conditions. [1]
1. To prevent **contamination by bacteria or fungi** (which can grow very rapidly on the nutrient media).
- (c) With reference to Table 3.1,
- (i) Describe the effects of auxin and cytokinin on callus growth. [3]
1. *Both hormones need to be present for any growth to occur.*
 2. *The type of growth depends on the relative concentrations of the auxin and cytokinin.*
 3. *A low level of cytokinin (0.02 mg dm⁻³) to auxin (2.00 mg dm⁻³) triggers root growth.*
 4. *Increasing the relative level of cytokinin (0.50 mg dm⁻³) to auxin (2.00 mg dm⁻³) triggers shoot growth.*
- (ii) Suggest what treatment should be given to the shoots developed from callus in treatment **D** to turn them into plantlets. [2]
1. *From the table, transferring the shoots from treatment D to treatment B /*
 2. *changing the cytokinin levels from 0.50 mg dm⁻³ to 0.02 mg dm⁻³ would stimulate the growth of roots and the developments of plantlets.*
- (d) Outline, using **one** named example, the process of genetic engineering that could increase crop yield. [4]
1. *Any named example, insect-resistant (Bt) / herbicide-tolerant crop (cotton, corn, soybean, tomato plant)*
- Process of genetic engineering:
2. *The gene for Bt toxin (from Bacillus thuringiensis) / the gene encoding glyphosate-resistance is isolated.*

3. The gene of interest is joined to a tumour-inducing / Ti plasmid that has been cut with the same restriction enzyme using DNA ligase.
4. The resultant recombinant plasmid is transformed into *Agrobacterium tumefaciens*.
5. Selection for *Agrobacterium* cells that have successfully taken up the recombinant plasmid, which are then cultured with leaf discs / callus tissue to mediate the transfer of the recombinant plasmid.
6. Alternatively, the gene of interest can be inserted into plant tissue using a gene gun.

Explanation how growing the genetically modified crop can lead to an increase in crop yield:

7. Bt crops are protected against damage by insect pests, so farmers have higher crop yield.
8. Farmers may apply a specific herbicide to control weeds population, without damaging their herbicide-tolerant crops. There will be a higher yield because of less competition from weeds.

(e) Identify **one** possible hazard and benefit of plant genetic engineering.

[2]

Possible hazard(s):

1. *Emergence of Bt toxin-resistant insects (by natural selection)*
2. *Genetically modified crop becomes a "weed" that out-competes wild crops for nutrients, space etc.*
3. *Possible transfer (via pollination) of the herbicide-resistant gene into weeds, creating weeds that are resistant to herbicides.*
4. *Accidental incorporation of a antibiotic selection marker into a pathogenic bacteria might result in antibiotic resistance of that bacterial species.*

Possible benefit(s):

1. *Economic benefit to farmers (either higher crop yield, lower output as no need to purchase pesticides / herbicides etc.)*
2. *Health benefit to consumers (because farmers use less pesticides).*

[Total: 15]

- 4 (a) Compare the mutations responsible for cystic fibrosis (CF), X-linked and ADA-deficient severe combined immunodeficiency (SCID) and comment on their pattern of inheritance.[10]

	X-linked SCID	ADA-deficiency SCID	Cystic fibrosis (CF)	Marks
Inheritance	X-linked	Autosomal recessive	Autosomal recessive	1
Chromosome affected	X chromosome	Chromosome 20 of ADA gene	Chromosome 7 of CFTR gene	1
Clinical symptoms	Absence of infection fighting cells i.e. T and B cells. Patients will be prone to virus and bacterial infection	Absence of infection fighting cells i.e. T and B cells. Patients will be prone to virus and bacterial infection	Accumulation of sticky mucus, leading to chronic infection	

Pattern of inheritance for X-linked SCID:

1. Thus, a female who is a carrier for X-linked SCID may show no signs of immunodeficiency
2. Her sons and daughters have a 50 percent chance of inheriting the X chromosome with the mutation
3. The son who has inherited one copy of the mutant X chromosome will have X-linked SCID
4. The daughter who has inherited one copy of the mutant X chromosome will be a carrier like their mother

Pattern of inheritance for ADA-deficient SCID:

5. Sufferers must be homozygotes since it is a recessive disorder
6. Meaning both parents (if carriers) must pass on the defective gene for any of their children to get the disease
7. Heterozygotes are carriers as they carry only one copy of the faulty gene
8. The chances of 2 carriers having a child with cystic fibrosis are 1 in 4

Pattern of inheritance for cystic fibrosis:

9. Sufferers must be homozygotes since it is a recessive disorder
10. Meaning both parents (if carriers) must pass on the defective gene for any of their children to get the disease
11. Heterozygotes are carriers as they carry only one copy of the faulty gene
12. The chances of 2 carriers having a child with cystic fibrosis are 1 in 4

- (b) With the recent food shortages world wide this year 2008, there has been a greater demand for alternative sources of food. Discuss some of the ethical and social implications of using genetically modified organisms to solve the demand for food. [10]

1. **Definition of GMO** -Genetically Modified Organism: Organism that has acquired one or more genes by artificial means. The genes may or may not be from the same species.

Social implications (max 4)

Threat to Human Safety

2. Genetic markers- **Vectors used for transforming plant/crop cells – contain antibiotic markers/ kanamycin resistance**; (example is Flavr Savr tomatoes)

Concern that the antibiotic resistant gene (if it is not properly broken down by the digestive system) may be passed to potentially harmful bacteria making them resistant to the antibiotics;

3. Allergens or Toxic/carcinogenic
 - Transgenic food can be potential sources of allergens.
 - Eg. those in peanuts can cause allergic reactions in some people.

Threat to Safety of the Environment

4. Genetically modified crops might establish themselves as weeds
Seeds from the transgenic crops that have been modified to **withstand unfavourable environmental conditions**, may be **carried by the wind** to other places. These crops grow quickly and may established themselves as weeds.
5. Spread of resistance from genetically modified crops to weeds
6. Pollen grains from genetically modified plants (eg. Herbicide resistant plant) can be carried in the wind and fertilise wild relatives and weeds. - This may result in "superweeds" – weeds that are **resistant to herbicide**. These may become more invasive (overrun agricultural areas rapidly) or **compete with the natural species**.

Upsetting the ecological balance

7. the genetically modified organisms (eg. Giant salmon) themselves may be more aggressive thus causing fear that they may upset the balance of the native species.
8. decrease in biodiversity;

Use of herbicide

9. another concern is that **developing herbicide-resistant plants** may encourage the use of greater amounts of herbicides;
10. the possibility of **pests developing resistance** to toxin produced by crop plants eg. Bt toxin;

Ethical implications (max of 5)

11. Is **genetic manipulation** of animals and plants in principle acceptable? Do we have the **right to tamper with nature by mixing genes among species**?
12. Is it acceptable to engineer any organism to be used as food for human consumption. - For example, the use of **BST in dairy cattle increases the risk of mastitis**. There seems to be little concern about whether the animals are **biologically 'designed' to withstand the additional burden of increased milk or egg or meat production**.
13. Religious concerns or dietary restrictions- Eg. Plants that have been modified to carry animal genes or vice versa;
14. Should anyone be **able to patent** a genetically modified animal or plant? - Many research companies have sought to patent the transgenic plants or animals that they have developed. Some people argued that patenting **animals is itself unethical as it reduces them to the level of objects**. If patenting is not allowed, how else could a company protect the results of their research program?
15. Is genetic engineering to make a staple crop more resist in marginal conditions (e.g. drought, cold) a potential boon for Third World agriculture, or another danger of increased dependence on rich "developed" countries? i.e. richer countries benefiting at the expense of the poorer countries;
16. world food production may be dominated by a small number of large companies with the technical know-how;
17. How great are the potential risks involved in releasing genetically modified organisms into the biosphere without knowing all the possible consequences?
18. Should products on sale be labeled to indicate that genetic engineering was involved in their production?